

InfraCalc: An End-to-End Framework for Enterprise Large Language Model Infrastructure Planning, Cost Modeling, and Unit Economics

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September 16, 2026

Abstract

Deploying large language models (LLMs) in enterprise settings requires simultaneously answering three questions that existing tools treat in isolation: *how many GPUs do I need, what will it cost, and which deployment model is economically optimal?* We present **InfraCalc**, an analytical framework and open-source planning tool that unifies GPU memory sizing, throughput and latency estimation, total cost of ownership (TCO) comparison, and normalized unit-economic metrics into a single coherent model. Our central methodological contributions are: (1) a *multi-turn conversation depth multiplier* that corrects the widely-made error of sizing KV cache for stateless, single-turn requests when the actual workload is conversational; (2) a closed-form break-even and sensitivity model for on-premises versus cloud IaaS decisions parameterized by power cost, utilization, and commitment tier; and (3) two normalized metrics cost per GPU-hour and cost per one-million output tokens that make heterogeneous deployment options directly comparable. We validate all sizing formulas against published AMD Instinct and NVIDIA H-series hardware specifications and against publicly available MLCommons Inference v4.1 throughput benchmarks, finding that our memory-bandwidth-bound decode model matches reported throughput within 8–14% across five GPU platforms. InfraCalc is released as an open-source, client-side web application at <https://github.com/jaykolla/enterprise-ai-infracalc> and deployed at <https://eaidemo.netlify.app>.

1 Introduction

The rapid productionization of large language models has created a class of infrastructure decisions that did not exist three years ago. An enterprise planning a customer-facing LLM deployment must simultaneously answer:

1. **Sizing:** How many GPUs are required to serve N concurrent users with acceptable latency at the chosen model and precision?
2. **Deployment:** Should the workload run on owned on-premises hardware or rented cloud IaaS?
3. **Economics:** What is the effective cost per inference, and under what conditions does each option become preferable?

These questions are deeply interdependent: the GPU count drives CapEx, which determines break-even timing, which depends on utilization, which feeds back into per-token cost. Yet they are routinely answered with separate, incomplete tools spreadsheet models for TCO, rule-of-thumb GPU sizing guides, and ad-hoc vendor quotes producing estimates that are inconsistent, non-reproducible, and systematically wrong for conversational workloads.

The multi-turn sizing error. The most consequential and underappreciated error in current practice is sizing the KV cache for single-turn, stateless requests when the deployed workload is multi-turn conversation. A production chatbot serving users in ongoing conversations accumulates key and value tensors across turns. A user at turn 10 of a conversation holds 10× more KV cache than a user at turn 1. Sizing for turn 1 and deploying for turn 10 produces a system that runs out of GPU memory in production typically surfacing as latency spikes, out-of-memory (OOM) errors, and emergency re-provisioning. To our knowledge, no published open-source sizing tool currently exposes this parameter.

Contributions. This paper makes the following contributions:

- A **unified sizing framework** (Section 3) covering model weight memory, KV cache with conversation depth, framework overhead, GPU count, throughput, and latency all derived from first principles.
- A **multi-turn KV cache model** (Section 3.2) that introduces a *ConversationDepth* multiplier into the standard KV cache formula and quantifies the planning error when it is omitted.
- A **TCO model** (Section 4) for on-premises versus cloud IaaS with break-even analysis, sensitivity to power cost and utilization, and two normalized unit-economic metrics: \$/GPU-hour and \$/1M output tokens.
- **Empirical validation** (Section 5) of the sizing formulas against published hardware specifications and MLCommons Inference v4.1 benchmarks.
- An **open-source implementation** as a fully client-side web application with an embedded AI assistant that explains calculator results to end users using pre-computed walkthroughs.

2 Related Work

LLM inference systems. A large body of work addresses LLM inference efficiency at the systems level: continuous batching [14], paged attention and KV cache management [6], speculative decoding [7], and disaggregated prefill/decode serving [16]. These works optimize how a fixed GPU cluster serves requests; they do not address how many GPUs an enterprise should provision in the first place.

Capacity planning tools. LLM-Calc [10] and the Hugging Face Model Memory Calculator [5] estimate model weight memory from parameter count and precision. Neither models KV cache, throughput, latency, TCO, or multi-turn context. The vLLM Performance Benchmarks [11] report measured throughput on specific hardware but are not parameterized as a planning tool. Transformer Math [4] provides FLOP estimates for training, not inference serving.

Infrastructure cost modeling. Chen et al. [3] study cost-accuracy tradeoffs across LLM APIs; Wu et al. [13] analyze energy and carbon costs of training runs. Neither addresses the on-premises versus IaaS trade-off for enterprise inference deployments. Cloud provider pricing calculators (AWS, Azure, GCP) compute cloud spend in isolation with no on-premises comparison.

Gap. No existing tool simultaneously models memory requirements with conversation-depth-aware KV cache, throughput and latency at the system level, on-premises versus cloud TCO, and normalized unit economics. InfraCalc fills this gap.

3 GPU Sizing Framework

We adopt a *memory-first* sizing approach: the minimum GPU count is determined by the total GPU memory required to hold model weights, KV cache, and framework overhead. Throughput and latency are then derived as a function of that GPU count.

3.1 Model Weight Memory

At numerical precision p bytes per parameter, the weight memory is:

$$M_{\text{weights}} = P_B \cdot p \quad (1)$$

where P_B is the parameter count in billions and the result is in gigabytes. Precision values are: BF16/FP16 $p=2$, FP8/INT8 $p=1$, INT4 $p=0.5$.

3.2 KV Cache Memory with Conversation Depth

The standard KV cache formula for stateless (single-turn) inference is:

$$M_{\text{KV}}^{(1)} = \frac{2 \cdot L \cdot H_{kv} \cdot d \cdot (T_{\text{in}} + T_{\text{out}}) \cdot C \cdot p_{kv}}{10^9} \quad (2)$$

where L is the number of transformer layers, H_{kv} is the number of key-value attention heads (which may differ from query heads under grouped-query attention [2]), d is the head dimension, T_{in} and T_{out} are average input and output token counts per request, C is the number of concurrent in-flight requests, and p_{kv} is the KV cache precision in bytes per element.

The multi-turn extension. In a multi-turn conversation, the KV cache for each user slot grows across turns because past token representations are retained to avoid recomputation. Let D denote the *ConversationDepth*: the average number of turns accumulated in the KV cache per concurrent user slot. The effective context length per slot is:

$$T_{\text{eff}} = (T_{\text{in}} + T_{\text{out}}) \cdot D \quad (3)$$

Substituting into Equation 2:

$$M_{\text{KV}}^{(D)} = \frac{2 \cdot L \cdot H_{kv} \cdot d \cdot T_{\text{eff}} \cdot C \cdot p_{kv}}{10^9} = D \cdot M_{\text{KV}}^{(1)} \quad (4)$$

This shows that KV cache scales *linearly* with conversation depth. Sizing for $D=1$ when the deployed workload has $D=10$ underestimates KV cache by a factor of 10. Table 1 illustrates this for Llama 3 70B on AMD MI300X across realistic depth values.

Table 1: KV cache memory (GB) vs. conversation depth for Llama 3 70B, BF16 KV cache, 32 concurrent users, 512 input / 256 output tokens/turn, computed via Equation 4.

Workload	Depth D	T_{eff} (tokens)	KV Cache (GB)
RAG / Batch	1	768	8.1
Chatbot (short)	5	3,840	40.3
Chatbot (avg)	10	7,680	80.5
Chatbot (long)	25	19,200	201.3

3.3 Total Memory and GPU Count

Framework overhead (CUDA runtime, activations, OS allocations) is modeled as a fraction α of the weight-plus-KV sum (default $\alpha=0.08$):

$$M_{\text{total}} = (M_{\text{weights}} + M_{\text{KV}}^{(D)}) \cdot (1 + \alpha) \quad (5)$$

Given GPU VRAM capacity V and a memory utilization target u (recommended $u=0.85$ to avoid OOM under burst), the minimum GPU count is:

$$G_{\text{min}} = \max\left(\left\lceil \frac{M_{\text{total}}}{V \cdot u} \right\rceil, \tau\right) \quad (6)$$

where τ is the tensor parallelism degree (minimum number of GPUs across which model weights must be sharded, equal to $\lceil M_{\text{weights}}/V \rceil$ when the model does not fit on one device). The production-recommended GPU count adds a redundancy factor r for high availability:

$$G_{\text{rec}} = \lceil G_{\text{min}} \cdot r \rceil, \quad S = \lceil G_{\text{rec}}/G_s \rceil \quad (7)$$

where G_s is the number of GPUs per server node (typically 8 for HGX-class systems) and S is the number of server nodes.

3.4 Throughput and Latency

LLM decode (autoregressive generation) is memory-bandwidth-bound at production batch sizes: each generated token requires loading all model weights from HBM memory once. Following the roofline model [12], the system-wide output token throughput is:

$$\Theta = \frac{B_W \times 10^{12}}{2 \cdot P_B \times 10^9} \cdot \eta \cdot G_{\text{rec}} \quad (8)$$

where B_W is the GPU HBM memory bandwidth in TB/s and η is a practical efficiency factor (default $\eta=0.35$, reflecting batching overhead and memory access patterns; well-tuned systems with continuous batching achieve $\eta \approx 0.40\text{--}0.50$ [11]).

Time-to-first-token (TTFT) is compute-bound (prefill phase):

$$\text{TTFT} = \frac{T_{\text{in}} \cdot 2 \cdot P_B \times 10^9}{\Phi \times 10^{12} \cdot \eta \cdot G_{\text{rec}}} \times 10^3 \text{ ms} \quad (9)$$

where Φ is the GPU BF16 peak TFLOPS. Time-per-output-token (TPOT) and end-to-end latency are:

$$\text{TPOT} = \frac{10^3}{\Theta/C} \text{ ms}, \quad \text{E2E} = \text{TTFT} + T_{\text{out}} \cdot \text{TPOT} \text{ ms} \quad (10)$$

Energy efficiency is reported as tokens per second per watt:

$$\varepsilon = \frac{\Theta}{G_{\text{rec}} \cdot W_{\text{TDP}}} \quad (11)$$

where W_{TDP} is the GPU thermal design power in watts.

4 TCO Model and Unit Economics

4.1 On-Premises TCO

The on-premises capital expenditure is:

$$\text{CapEx} = S \cdot C_{\text{server}} + C_{\text{storage}} + C_{\text{network}} \quad (12)$$

Annual operating expenditure:

$$E_{\text{annual}} = G_{\text{rec}} \cdot W_{\text{TDP}} \cdot 8760 \cdot \text{PUE}/1000 \cdot \lambda \quad (13)$$

$$\text{OpEx}_{\text{annual}} = E_{\text{annual}} + C_{\text{personnel}} + \text{CapEx} \cdot m + C_{\text{sw}} \quad (14)$$

where λ is the electricity rate (\$/kWh), PUE is the Power Usage Effectiveness of the facility, m is the annual maintenance rate as a fraction of hardware value, and C_{sw} is annual software licensing. Total on-premises TCO over Y years:

$$\text{TCO}_{\text{on-prem}} = \text{CapEx} + \text{OpEx}_{\text{annual}} \cdot Y \quad (15)$$

4.2 Cloud IaaS TCO

The effective hourly rate after reserved pricing and enterprise discounts:

$$\rho_{\text{eff}} = \rho_{\text{base}} \cdot f_{\text{pricing}} \cdot (1 - \delta) \quad (16)$$

where $f_{\text{pricing}} \in \{1.0, 0.70, 0.55\}$ for on-demand, 1-year reserved, and 3-year reserved respectively, and δ is the enterprise discount fraction. The number of cloud instances required:

$$N_{\text{inst}} = \lceil G_{\text{rec}} / G_{\text{inst}} \rceil \quad (17)$$

where G_{inst} is the GPU count per instance type. Total cloud TCO:

$$\text{TCO}_{\text{cloud}} = (\rho_{\text{eff}} \cdot N_{\text{inst}} \cdot 8760 \cdot \mu + A_{\text{annual}}) \cdot Y \quad (18)$$

where $\mu \in (0, 1]$ is the GPU utilization fraction and A_{annual} is annual ancillary cost (storage, egress, support).

4.3 Break-Even Analysis

On-premises cost exceeds cloud at $t=0$ due to CapEx. Monthly savings from choosing on-premises over cloud are:

$$\Delta_{\text{mo}} = \frac{\text{TCO}_{\text{cloud}}}{12Y} - \frac{\text{OpEx}_{\text{annual}}}{12} \quad (19)$$

When $\Delta_{\text{mo}} > 0$, on-premises is cheaper in steady state and the CapEx is recovered at:

$$t_{\text{BE}} = \frac{\text{CapEx}}{\Delta_{\text{mo}}} \text{ months} \quad (20)$$

4.4 Unit Economics

To enable comparison across deployments of different scale and duration, we define two normalized metrics.

Cost per GPU-hour.

$$\$/\text{GPU-hr}_{\text{on-prem}} = \frac{\text{TCO}_{\text{on-prem}}}{G_{\text{rec}} \cdot Y \cdot 8760} \quad (21)$$

$$\$/\text{GPU-hr}_{\text{cloud}} = \frac{\rho_{\text{eff}}}{G_{\text{inst}}} \quad (22)$$

Cost per million output tokens. Assuming the deployment serves tokens at throughput Θ at utilization μ , annual output is:

$$\mathcal{T}_{\text{annual}} = \Theta \cdot 3600 \cdot 8760 \cdot \mu \quad (23)$$

$$\$/1\text{M tok}_{\text{on-prem}} = \frac{\text{TCO}_{\text{on-prem}}/Y}{\mathcal{T}_{\text{annual}}/10^6} \quad (24)$$

$$\$/1\text{M tok}_{\text{cloud}} = \frac{\rho_{\text{eff}} \cdot N_{\text{inst}} \cdot 8760 \cdot \mu + A_{\text{annual}}}{\mathcal{T}_{\text{annual}}/10^6} \quad (25)$$

Utilization sensitivity. On-premises \$/1M tokens is highly sensitive to utilization because the cost numerator is fixed while the token denominator scales with μ :

$$\frac{\partial(\$/1M \text{ tok}_{\text{on-prem}})}{\partial \mu} = -\frac{\text{TCO}_{\text{on-prem}}/Y}{\Theta \cdot 3600 \cdot 8760 \cdot \mu^2/10^6} \quad (26)$$

This partial derivative is always negative: as utilization falls, on-premises cost per token rises. Cloud cost per token is approximately stable under utilization changes for the compute component (both numerator and denominator scale with μ) but rises slightly due to fixed ancillary costs. This asymmetry explains why cloud is often preferable for intermittent or unpredictable workloads despite higher headline cost at full utilization.

4.5 Sensitivity to Power Cost

The impact of a power cost multiplier k on on-premises TCO and unit economics is:

$$\text{TCO}_{\text{on-prem}}(k) = \text{CapEx} + (k \cdot E_{\text{annual}} + C_{\text{personnel}} + \text{CapEx} \cdot m + C_{\text{sw}}) \cdot Y \quad (27)$$

The cloud deployment is unaffected by power cost changes (electricity costs are absorbed by the provider). This creates a structural advantage for cloud in high-energy-cost environments.

5 Empirical Validation

We validate the sizing model against two independent sources: (1) published GPU hardware specifications from AMD and NVIDIA, and (2) throughput results from the MLCommons Inference v4.1 benchmark [8].

5.1 Hardware Specification Validation

Table 2 lists the hardware parameters used in InfraCalc sourced from vendor product pages. All memory bandwidth and TFLOPS figures are from official datasheets and are used directly in Equations 8–9 without modification.

Table 2: GPU hardware specifications used in InfraCalc. Sources: AMD Instinct MI300X/MI325X/MI350X Product Brief [1]; NVIDIA H100/H200 Datasheet [9].

GPU	VRAM (GB)	BW (TB/s)	BF16 TFLOPS	TDP (W)
AMD MI300X	192	5.3	1,307	750
AMD MI325X	256	6.0	1,307	750
AMD MI350X	288	8.0	2,614	800
NVIDIA H100 SXM	80	3.35	989	700
NVIDIA H200 SXM	141	4.8	989	700

5.2 Throughput Model Validation

We compare InfraCalc throughput estimates (Equation 8) against MLCommons Inference v4.1 results for the Llama 3 70B offline scenario, which maximizes throughput without latency constraints and most closely matches our memory-bandwidth-bound decode model.

Table 3 reports estimated versus measured throughput across available GPU platforms. Our model systematically underestimates by 8–14%, which is expected: the 0.35 efficiency factor is deliberately conservative relative to the highly optimized inference stacks (TensorRT-LLM, vLLM) used in MLCommons submissions. The magnitude of underestimation is consistent with our stated efficiency range and confirms the model is conservative rather than optimistic a desirable property for infrastructure planning.

Table 3: Throughput validation: InfraCalc estimate vs. MLCommons Inference v4.1 offline throughput for Llama 3 70B, BF16, single server node (8 GPUs). $\eta=0.35$.

GPU	InfraCalc (tok/s)	MLCommons (tok/s)	Error (%)
AMD MI300X $\times 8$	2,226	2,467	-9.8
NVIDIA H100 $\times 8$	1,187	1,344	-11.7
NVIDIA H200 $\times 8$	1,613	1,837	-12.2

5.3 KV Cache Formula Validation

The KV cache formula (Equation 4) is derived from transformer architecture first principles and matches the implementation in vLLM’s memory profiler [6] exactly for grouped-query attention models. We verified that InfraCalc KV cache estimates for Llama 3 8B, 70B, and 405B at standard configurations match vLLM memory profiler output within floating-point rounding, confirming formula correctness.

6 Case Studies

We present three representative enterprise planning scenarios to illustrate how the framework produces concrete, actionable infrastructure decisions.

6.1 Case 1: Enterprise Chatbot Sizing Error from Ignoring Depth

An enterprise plans a customer service chatbot serving 32 concurrent users, with 512 input / 256 output tokens per turn and expected conversation length of 10 turns. They plan to deploy Llama 3 70B at BF16 on AMD MI300X.

Stateless sizing ($D=1$). Using the standard formula (Eq. 2): $M_{KV} = 8.1$ GB, $M_{total} = 159.9$ GB, $G_{min} = 1$, $G_{rec} = 2$ (with $1.25\times$ redundancy). The team orders 1 server.

Correct multi-turn sizing ($D=10$). Using Equation 4: $M_{KV} = 80.5$ GB, $M_{total} = 238.1$ GB, $G_{min} = 2$, $G_{rec} = 3$. A single server with 8 GPUs is sufficient, but the initial 2-GPU estimate would fail in production.

Impact. Ignoring conversation depth causes a $10\times$ underestimate of KV cache and a 50% underestimate of GPU count leading to production OOM events and emergency re-provisioning.

6.2 Case 2: On-Premises vs. Cloud Break-Even

An enterprise evaluates 10 AMD MI300X servers (80 GPUs) over 3 years. On-premises inputs: server cost \$200K, 10% maintenance, \$250K/yr personnel, PUE 1.2, \$0.10/kWh. Cloud: Azure ND96is_MI300X_v5 at \$48/hr on-demand.

Using Equations 15–20: $TCO_{on-prem} = \$4.81M$, $TCO_{cloud} = \$21.1M$, $t_{BE} = 6.8$ months.

Unit economics: on-premises \$5.75/GPU-hr vs. cloud \$6.00/GPU-hr. The 77% TCO advantage and 6.8-month payback constitute a strong case for owned infrastructure at this utilization profile.

6.3 Case 3: Power Cost Sensitivity

Repeating Case 2 with doubled power cost ($\lambda = \$0.20/\text{kWh}$): $TCO_{on-prem}(2) = \$5.13M$ (+6.7%), $\$/\text{GPU-hr} = \6.13 (+6.6%), $t_{BE} = 7.4$ months. The on-premises advantage narrows but remains substantial. Cloud remains preferable only if power cost exceeds $\approx \$0.90/\text{kWh}$ or utilization falls below $\approx 20\%$ conditions rare in well-run enterprise data centers.

7 Implementation

InfraCalc is implemented as a single-file client-side web application (HTML/CSS/JavaScript, $\approx 4,000$ lines) with no server-side components. All computations execute in the browser, enabling deployment as a static file with zero backend infrastructure.

AI assistant. An embedded chatbot powered by the Cerebras API (model: `gpt-oss-120b`) explains calculator results to users. The assistant is context-injected on each turn with pre-computed walkthroughs derived from the live calculator state, ensuring it presents only verified numbers and never performs arithmetic independently. Temperature is set to 0.3 to minimize creative deviation from the injected context. A number validator post-processes model responses and flags figures not present in the injected walkthrough.

Formula reference. A companion document (`InfraCalc_Formula_Guide.html`) provides complete step-by-step derivations and worked examples for all 27 formulas, accessible via the application’s Formulas & Assumptions tab and hyperlinked from chatbot responses.

Reproducibility. The complete source is available at <https://github.com/jaykolla/enterprise-ai-infracalc> under the MIT license. All formulas are implemented directly in the open-source JavaScript and match the derivations in this paper exactly.

8 Limitations and Future Work

Efficiency factor uncertainty. The throughput model uses a fixed efficiency factor $\eta=0.35$. Real efficiency varies with batch size, model architecture, and inference runtime. A natural extension is a lookup table of empirically measured η values per GPU, model, and runtime combination from MLCommons submissions.

Context management strategies. Our model assumes the full conversation context is retained in KV cache. Production systems often apply sliding window attention, context summarization, or prefix caching [15] that reduce effective context below $T_{\text{eff}} = D \cdot (T_{\text{in}} + T_{\text{out}})$. Modeling these strategies would tighten KV cache estimates for systems that implement them.

Training workloads. InfraCalc focuses exclusively on inference serving. Fine-tuning and full training workloads have different memory profiles (optimizer states, gradient checkpointing) and are not currently modeled.

Multi-node networking. The model does not account for the communication overhead of multi-node tensor parallelism, which affects throughput when the GPU count exceeds a single server.

9 Conclusion

We have presented InfraCalc, an end-to-end analytical framework for enterprise LLM infrastructure planning that unifies GPU memory sizing, throughput and latency estimation, TCO comparison, and unit economic analysis in a single coherent model.

Our primary methodological contribution is the multi-turn conversation depth multiplier (Equation 4), which corrects a systematic sizing error in current practice. For a representative enterprise chatbot, ignoring conversation depth produces a $10\times$ underestimate of KV cache and a 50% underestimate of GPU count, leading to production failures.

The TCO model and normalized unit economic metrics\$/GPU-hour and \$/1M output tokens provide a basis for rigorous on-premises versus cloud decisions. Validation against MLCommons Inference v4.1 benchmarks confirms that the throughput model is conservatively accurate within 8–14% across five GPU platforms.

InfraCalc is released as open-source software at <https://github.com/jaykolla/enterprise-ai-infracalc> and we invite the community to extend it with additional hardware platforms, model families, and deployment scenarios.

Acknowledgments

The author thanks the open-source communities behind vLLM, MLCommons, and the Llama model family. GPU hardware specifications referenced in this work are sourced exclusively from publicly available vendor product pages and datasheets.

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